

Republic F-105D Thunderchief

Famous for their WWII Thunderbolt, Republic built the largest single-seat, single-engine aircraft ever in this long-range tactical fighter-bomber. Entering service in 1959, the aircraft carried out more air strikes over North Vietnam between 1966 and 1971 than any other USAF aircraft. Losses were heavy and since it was expensive to maintain, it was gradually phased out.



ENGINE 24,500lb (11,110kg) thrust P&W J75-P-19W turbojet
WINGSPAN 34ft 11in (10.7m) **LENGTH** 64ft 3in (19.6m)
TOP SPEED 1,390mph (2,237kph) **CREW** 1
ARMAMENT 1 x 20mm M-61 Vulcan rotary cannon; 3,630kg (8,000lb) bombload, 6,000lb (2,720kg) missiles on wing pylons

Vought A-7E Corsair II



Derived from but completely different to the Crusader, this attack aircraft sacrificed supersonic speed for carrying capacity and range. Its stubby shape earned it the nickname SLUF – “Short Little Ugly Fella.” Ordered by the US Navy and Air Force in 1966, it was still in Navy frontline service during the Gulf War.

ENGINE 15,000lb (6,800kg) thrust Allison TF41-A-2 turbofan
WINGSPAN 38ft 9in (11.8m) **LENGTH** 46ft 1in (14.1m)
TOP SPEED 698mph (1,123kph) **CREW** 1
ARMAMENT 1 x 20mm M-61 Vulcan rotary cannon, up to 15,000lb (6,800kg) bombs, rockets, or missiles

Vought (F-8A) F8U-1 Crusader



The US Navy's main fleet defense fighter from 1957, the Crusader (French F-8E type shown) outperformed the F-100 using the same engine. The high-mounted wing tilted up to allow slower landing and takeoff on carrier decks. Effective in combat, the F-8 was gradually replaced by the better (but more expensive) Phantom.

ENGINE 16,200lb (7,330kg) thrust P&W J57-P-4 turbojet
WINGSPAN 35ft 8in (10.9m) **LENGTH** 54ft 3in (16.5m)
TOP SPEED 1,013mph (1,630kph) **CREW** 1
ARMAMENT 4 x 20mm cannon; 2 x Sidewinder air-to-air missiles, 32 rockets in belly pack

deployed to immense destructive effect in Korea without exciting notable unease in the United States. But in Vietnam, the more successful aircraft were in achieving destruction, the more criticism they provoked. Standards of tolerance for civilian casualties were falling, and especially, it seemed, for civilian casualties caused by air attack. This was a potentially major inhibition on the use of aircraft in what might be termed “wars in peacetime.”

The Six-Day War

While the Americans were unhappily embroiled in Vietnam, in 1967 the Israelis gave an exceptional demonstration of the incisive use of air power in their Six-Day War with their Arab neighbors in the Middle East. The Israeli Air Force (IAF) was a self-conscious elite with only a few hundred combat aircraft. These were exclusively French – mostly Dassault Mirage IIIs, Mystères, Super Mystères, and Ouragans – and most were fighter-bombers, since Israel did not have the budget for a separate fighter and bomber force. For years the IAF had planned to win air superiority at the very start of a war by destroying enemy aircraft on the ground. It would then use its command of the air to provide ground forces with decisive support. This was a strategy with clear and achievable objectives, requiring complete ruthlessness in disregard for such niceties as a prior declaration of war.

Early on the morning of June 5, 1967, 196 Israeli aircraft, in flights of four, headed at low altitude across the Mediterranean and the Sinai desert to deliver coordinated strikes against 10 airfields in Egypt. It was the most effective surprise air attack since Pearl Harbor. The Israeli pilots maintained complete radio silence, and the aircraft were not picked up by Egyptian radar. The Israeli fighter-bombers found Egyptian

aircraft lined up on their hard-stands. They made a first pass with bombs and a second pass strafing Egyptian airplanes on the ground. Some airplanes were armed with specialized runway-cratering bombs, which were rocket-propelled into the runway's concrete surface and exploded, creating holes about 6ft (2m) deep.

Most of the Egyptians' 600 Soviet-supplied aircraft were destroyed on the ground. Those that managed to engage the Israelis in air combat were outfought and shot down in large numbers. The low level of the attack went under surface-to-air missile (SAM) radars and the missiles were wholly ineffectual, although anti-aircraft fire claimed a score of Israeli victims. Once the Egyptians were out of action, the Israelis went on to attack airfields in Iraq, Syria, and Jordan. Complete air superiority was achieved in 10 hours. With command of the air, over the following days the IAF attacked Arab armored columns and artillery positions to devastating effect.

Israeli success

The Six-Day War made the Mirage III momentarily the most famous aircraft in the world, but the success of the Israelis was not due to technological superiority. It was partly a demonstration of the gulf that could exist in air combat between pilots selected and trained to be outstanding fighting men and merely competent adversaries. It was also a sign of how much more effective air power could be in a theater of war in which the weather was usually clear, the terrain left armies exposed and without the possibility of concealment, and much of the fighting took place in areas with little or no civilian population. And it showed how the military effectiveness of aircraft could benefit from the absence of politically imposed rules or limits.

MARCEL DASSAULT

MARCEL DASSAULT (1892–1986) was the French airplane maker responsible for the Mirage and Mystère fighter-bombers. Born Marcel Bloch, he was one of the first Frenchmen to take a degree in aeronautical engineering. During World War I, in partnership with Henri Potez, he produced a propeller for the famous SPAD fighters and set up the aircraft-manufacturing company, SEA. This went bust after the war and Bloch turned to making furniture. In the 1930s he set up Avions Marcel Bloch to build all-metal aircraft.

As a prominent Jewish businessman, Bloch's position after the Nazi victory was precarious. He joined the French Resistance, but was arrested by the Gestapo and sent to Buchenwald concentration camp. He survived this ordeal and after the war changed his name to Dassault.

Dassault built up a postwar business empire centered on aviation. The outstanding jet aircraft his company produced kept France in the fore of military aviation development. He died shortly after the first flight of the Rafale fighter.

LONG CAREER

The long career of French aviation pioneer Marcel Dassault (originally Bloch) stretched from the days of wire-and-strut biplanes to the era of supersonic jets.

